Informe de evidencias
Laboratorio de contenedores Docker

Asignatura / práctica	Laboratorio de contenedores Docker
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Este informe organiza las evidencias visuales por bloques del laboratorio. Se incluyen únicamente las actividades que cuentan con captura adjunta y, al final, se listan las actividades del PDF que no quedaron evidenciadas en el material entregado.

1. Alcance del informe

La guía del laboratorio contempla actividades de preparación del entorno, manejo de imágenes, contenedores, volúmenes, redes, Docker Compose y prueba de instrucciones en Dockerfile. En este documento se consolida la evidencia disponible a partir de las capturas entregadas y se relaciona cada una con el paso correspondiente del laboratorio.

El objetivo es demostrar, de forma ordenada y verificable, el avance logrado en la ejecución de la práctica y facilitar la revisión por parte del docente.

2. Resumen de evidencias disponibles

Bloque	Actividad cubierta	Estado	Figuras
0. Preparación	Verificación de instalación y del daemon de Docker.	**Evidenciado**	Figuras 1-2
1. Docker Image	Pull, build, ejecución de imagen, etiquetado, inspect, save/load y eliminación de etiqueta.	**Evidenciado**	Figuras 3-9
2. Docker Container	Creación de contenedores, puertos, listado, inspect, stop/start, modo interactivo, pause/unpause y eliminación.	**Evidenciado**	Figuras 10-15
3. Volúmenes	Listado/inspect del volumen y uso del volumen con Nginx.	**Evidenciado**	Figuras 16-17
4. Redes	Listado, creación, inspección, conectividad y desconexión de la red bridge propia.	**Evidenciado**	Figuras 18-20
5. Docker Compose	Creación/corrección del compose, up, ps, curl, logs, down y escalado.	**Evidenciado**	Figuras 21-24
6. Dockerfile (instrucciones)	Prueba completa del ejemplo corto con archivo.txt e inspección de label/env/cmd/volume.	**Evidenciado**	Figuras 25-26

Con las evidencias complementarias incorporadas, el laboratorio queda documentado en todos sus bloques principales.

3. Desarrollo de evidencias

A continuación se presenta el detalle de las capturas organizadas por bloques temáticos. Cada figura conserva el orden cronológico de la práctica y describe el resultado visible en la consola.

0. Preparación

Se observa que Docker responde correctamente mostrando información del cliente y del servidor, así como los datos generales del motor.

```
28 de mar 16:54
chaya@chaya-HP-240-G8-Notebook-PC: ~
chaya@chaya-HP-240-G8-Notebook-PC:~$ ^[[200~docker version
docker: no se encontró la orden
chaya@chaya-HP-240-G8-Notebook-PC:~$ docker version
Client: Docker Engine - Community
 Version:           29.3.0
 API version:       1.54
 Go version:        go1.25.7
 Git commit:        5927d80
 Built:             Thu Mar  5 14:26:07 2026
 OS/Arch:           linux/amd64
 Context:           default

Server: Docker Engine - Community
 Engine:
  Version:          29.3.0
  API version:      1.54 (minimum version 1.40)
  Go version:       go1.25.7
  Git commit:       83bca51
  Built:            Thu Mar  5 14:26:07 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.2.2
  GitCommit:        301b2dac98f15c27117da5c8af12118a041a31d9
 runc:
  Version:          1.3.4
  GitCommit:        v1.3.4-0-gd6d73eb8
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
chaya@chaya-HP-240-G8-Notebook-PC:~$ docker info
Client: Docker Engine - Community
 Version:    29.3.0
 Context:    default
 Debug Mode: false
 Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.31.1
```

Figura 1. Verificación inicial de Docker mediante los comandos docker version y docker info.

0. Preparación / 1.1

La captura evidencia que el daemon está activo y que se inicia la descarga de la imagen nginx:latest desde Docker Hub.

```
28 de mar 17:24
chaya@chaya-HP-240-G8-Notebook-PC: ~
  /etc/cdi
  /var/run/cdi
 Swarm: inactive
 Runtimes: io.containerd.runc.v2 runc
 Default Runtime: runc
 Init Binary: docker-init
 containerd version: 301b2dac98f15c27117da5c8af12118a041a31d9
 runc version: v1.3.4-0-gd6d73eb8
 init version: de40ad0
 Security Options:
  apparmor
  seccomp
   Profile: builtin
  cgroupns
 Kernel Version: 6.17.0-14-generic
 Operating System: Ubuntu 25.10
 OSType: linux
 Architecture: x86_64
 CPUs: 4
 Total Memory: 7.022GiB
 Name: chaya-HP-240-G8-Notebook-PC
 ID: b0a4981d-125f-477b-b38c-e366b432a79e
 Docker Root Dir: /var/lib/docker
 Debug Mode: false
 Experimental: false
 Insecure Registries:
  ::1/128
  127.0.0.0/8
 Live Restore Enabled: false
 Firewall Backend: iptables

chaya@chaya-HP-240-G8-Notebook-PC:~$ docker pull nginx:latest
latest: Pulling from library/nginx
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
chaya@chaya-HP-240-G8-Notebook-PC:~$
```

Figura 2. Continuación de docker info y ejecución del comando docker pull nginx:latest.

1.1 Descargar imagen

Se confirma que la imagen nginx:latest queda disponible localmente en el host.

```
28 de mar  17:29
chaya@chaya-HP-240-G8-Notebook-PC: ~
Init Binary: docker-init
containerd version: 301b2dac98f15c27117da5c8af12118a041a31d9
runc version: v1.3.4-0-gd6d73eb8
init version: de40ad0
Security Options:
 apparmor
 seccomp
  Profile: builtin
 cgroupns
Kernel Version: 6.17.0-14-generic
Operating System: Ubuntu 25.10
OSType: linux
Architecture: x86_64
CPUs: 4
Total Memory: 7.022GiB
Name: chaya-HP-240-G8-Notebook-PC
ID: b0a4981d-125f-477b-b38c-e366b432a79e
Docker Root Dir: /var/lib/docker
Debug Mode: false
Experimental: false
Insecure Registries:
 ::1/128
 127.0.0.0/8
Live Restore Enabled: false
Firewall Backend: iptables

chaya@chaya-HP-240-G8-Notebook-PC:~$ docker pull nginx:latest
latest: Pulling from library/nginx
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
chaya@chaya-HP-240-G8-Notebook-PC:~$ docker images
                                                                                  1 Info →  U  In Use
IMAGE               ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest  452a468a4bf9       25.9kB         9.49kB    U
nginx:latest        7150b3a39203        240MB         65.8MB    U
chaya@chaya-HP-240-G8-Notebook-PC:~$
```

Figura 3. Validación del pull de nginx mediante el listado docker images.

1.2 Construir imagen desde Dockerfile

La evidencia muestra la ejecución satisfactoria de docker build -t miapp:v1 . dentro del directorio de trabajo.

```
28 de mar  17:52
chaya@chaya-HP-240-G8-Notebook-PC: ~/docker-lab-chayanne
~/docker-lab-chayanne
                                                                                  1 Info →  U  In Use
IMAGE               ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest  452a468a4bf9       25.9kB         9.49kB    U
nginx:latest        7150b3a39203        240MB         65.8MB    U
chaya@chaya-HP-240-G8-Notebook-PC:~$ mkdir docker-lab-chayanne
chaya@chaya-HP-240-G8-Notebook-PC:~$ ^C
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd ^[[200~mkdir docker-lab
bash: cd: demasiados argumentos
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd mkdir docker-lab
bash: cd: demasiados argumentos
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd mkdir docker-lab-chayanne
bash: cd: demasiados argumentos
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd docker-lab-chayanne
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ touch Dockerfile
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ nano Dockerfile
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker build -t miapp:v1 .
[+] Building 28.1s (7/7) FINISHED                                                  docker:default
 => [internal] load build definition from Dockerfile                                          0.7s
 => => transferring dockerfile: 148B                                                          0.0s
 => [internal] load metadata for docker.io/library/alpine:3.20                                5.7s
 => [internal] load .dockerignore                                                             0.4s
 => => transferring context: 2B                                                               0.0s
 => [1/3] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805  3.8s
 => => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805  0.2s
 => => sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd 3.63MB / 3.63MB  2.1s
 => => extracting sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd     0.6s
 => [2/3] WORKDIR /app                                                                        1.5s
 => [3/3] RUN echo "Hola desde Docker" > /app/mensaje.txt                                     4.3s
 => exporting to image                                                                        7.9s
 => => exporting layers                                                                       3.4s
 => => exporting manifest sha256:111327af693e1b839ebfff2cfeff3848f33bea86958c871b5224daf4f7a31c58  1.2s
 => => exporting config sha256:0ec7af156d37b75a216866ba4aa0b26c9e8a7327e59d5f4fdd9befedb8766ecb  0.4s
 => => exporting attestation manifest sha256:3eba7b941c13f6206d51e7075b5c33ab14113ce00074516aca08d8392b113045  1.7s
 => => exporting manifest list sha256:9e9d6f7ab66069e087198d247daa401b1eef78c4f7c3e22301fbee0fcec9a771  0.3s
 => => naming to docker.io/library/miapp:v1                                                   0.0s
 => => unpacking to docker.io/library/miapp:v1                                                0.5s
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 4. Construcción de la imagen miapp:v1 a partir de un Dockerfile basado en Alpine.

La salida confirma la existencia local de la imagen miapp:v1.

```
28 de mar  17:54
chaya@chaya-HP-240-G8-Notebook-PC: ~/docker-lab-chayanne
~/docker-lab-chayanne
nginx:latest         7150b3a39203        240MB         65.8MB    U
chaya@chaya-HP-240-G8-Notebook-PC:~$ mkdir docker-lab-chayanne
chaya@chaya-HP-240-G8-Notebook-PC:~$ ^C
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd ^[[200~mkdir docker-lab
bash: cd: demasiados argumentos
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd mkdir docker-lab
bash: cd: demasiados argumentos
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd mkdir docker-lab-chayanne
bash: cd: demasiados argumentos
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd docker-lab-chayanne
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ touch Dockerfile
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ nano Dockerfile
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker build -t miapp:v1 .
[+] Building 28.1s (7/7) FINISHED                                                                   docker:default
 => [internal] load build definition from Dockerfile                                                          0.7s
 => => transferring dockerfile: 148B                                                                          0.0s
 => [internal] load metadata for docker.io/library/alpine:3.20                                                5.7s
 => [internal] load .dockerignore                                                                             0.4s
 => => transferring context: 2B                                                                               0.0s
 => [1/3] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805  3.8s
 => => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805  0.2s
 => => sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd 3.63MB / 3.63MB                2.1s
 => => extracting sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd                     0.6s
 => [2/3] WORKDIR /app                                                                                        1.5s
 => [3/3] RUN echo "Hola desde Docker" > /app/mensaje.txt                                                     4.3s
 => exporting to image                                                                                        7.9s
 => => exporting layers                                                                                       3.4s
 => => exporting manifest sha256:111327af693e1b839ebfff2cfeff3848f33bea86958c871b5224daf4f7a31c58             1.2s
 => => exporting config sha256:0ec7af156d37b75a216866ba4aa0b26c9e8a7327e59d5f4fdd9befedb8766ecb               0.4s
 => => exporting attestation manifest sha256:3eba7b941c13f6206d51e7075b5c33ab14113ce00074516aca08d8392b113045  1.7s
 => => exporting manifest list sha256:9e9d6f7ab66069e087198d247daa401b1eef78c4f7c3e22301fbee0fcec9a771        0.3s
 => => naming to docker.io/library/miapp:v1                                                                   0.0s
 => => unpacking to docker.io/library/miapp:v1                                                                0.5s
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:v1             9e9d6f7ab660       12.1MB         3.63MB
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 5. Verificación de la imagen creada con docker images | grep miapp.

El contenedor imprime el mensaje esperado "Hola desde Docker", validando el CMD definido en el Dockerfile.

```
28 de mar  17:55
chaya@chaya-HP-240-G8-Notebook-PC: ~/docker-lab-chayanne
~/docker-lab-chayanne
chaya@chaya-HP-240-G8-Notebook-PC:~$ ^C
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd ^[[200~mkdir docker-lab
bash: cd: demasiados argumentos
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd mkdir docker-lab
bash: cd: demasiados argumentos
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd mkdir docker-lab-chayanne
bash: cd: demasiados argumentos
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd docker-lab-chayanne
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ touch Dockerfile
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ nano Dockerfile
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker build -t miapp:v1 .
[+] Building 28.1s (7/7) FINISHED                                                                   docker:default
 => [internal] load build definition from Dockerfile                                                          0.7s
 => => transferring dockerfile: 148B                                                                          0.0s
 => [internal] load metadata for docker.io/library/alpine:3.20                                                5.7s
 => [internal] load .dockerignore                                                                             0.4s
 => => transferring context: 2B                                                                               0.0s
 => [1/3] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805  3.8s
 => => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805  0.2s
 => => sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd 3.63MB / 3.63MB                2.1s
 => => extracting sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd                     0.6s
 => [2/3] WORKDIR /app                                                                                        1.5s
 => [3/3] RUN echo "Hola desde Docker" > /app/mensaje.txt                                                     4.3s
 => exporting to image                                                                                        7.9s
 => => exporting layers                                                                                       3.4s
 => => exporting manifest sha256:111327af693e1b839ebfff2cfeff3848f33bea86958c871b5224daf4f7a31c58             1.2s
 => => exporting config sha256:0ec7af156d37b75a216866ba4aa0b26c9e8a7327e59d5f4fdd9befedb8766ecb               0.4s
 => => exporting attestation manifest sha256:3eba7b941c13f6206d51e7075b5c33ab14113ce00074516aca08d8392b113045  1.7s
 => => exporting manifest list sha256:9e9d6f7ab66069e087198d247daa401b1eef78c4f7c3e22301fbee0fcec9a771        0.3s
 => => naming to docker.io/library/miapp:v1                                                                   0.0s
 => => unpacking to docker.io/library/miapp:v1                                                                0.5s
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:v1             9e9d6f7ab660       12.1MB         3.63MB
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker run --rm miapp:v1
Hola desde Docker
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 6. Ejecución de la imagen miapp:v1 con docker run --rm miapp:v1.

1.3 Etiquetar imagen

La verificación con docker images | grep miapp demuestra la presencia de las etiquetas miapp:v1 y miapp:latest.

```
28 de mar 18:04
chaya@chaya-HP-240-G8-Notebook-PC: ~/docker-lab-chayanne
~/docker-lab-chayanne

Hola desde Docker
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ nano DOckerfile
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:v1            9e9d6f7ab660       12.1MB         3.63MB
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker run --rm miapp:v1
Hola desde Docker
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker build -t miapp:v1 .
[+] Building 18.3s (7/7) FINISHED                                                                                docker:default
 => [internal] load build definition from Dockerfile                                                                      0.1s
 => => transferring dockerfile: 148B                                                                                      0.0s
 => [internal] load metadata for docker.io/library/alpine:3.20                                                           12.7s
 => [internal] load .dockerignore                                                                                         0.1s
 => => transferring context: 2B                                                                                           0.0s
 => [1/3] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805       1.1s
 => => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805       0.3s
 => CACHED [2/3] WORKDIR /app                                                                                             0.0s
 => CACHED [3/3] RUN echo "Hola desde Docker" > /app/mensaje.txt                                                          0.0s
 => exporting to image                                                                                                    1.2s
 => => exporting layers                                                                                                   0.0s
 => => exporting manifest sha256:111327af693e1b839ebfff2cfeff3848f33bea86958c871b5224daf4f7a31c58                         0.2s
 => => exporting config sha256:0ec7af156d37b75a216866ba4aa0b26c9e8a7327e59d5f4fdd9befedb8766ecb                           0.0s
 => => exporting attestation manifest sha256:b07c8b99befcf34e32295b392653ca377b68ad277ef27d72f0cc13e3dfb3037b             0.4s
 => => exporting manifest list sha256:67f2db8020c575ff247f8217f82eb50b5cfbcb95e4d4492087e724f6f52d761c                    0.2s
 => => naming to docker.io/library/miapp:v1                                                                               0.0s
 => => unpacking to docker.io/library/miapp:v1                                                                            0.1s
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:v1            67f2db8020c5       12.1MB         3.63MB
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker run --rm miapp:v1
Hola desde Docker
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker tag miapp:v1 miapp:latest
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:latest        67f2db8020c5       12.1MB         3.63MB
miapp:v1            67f2db8020c5       12.1MB         3.63MB
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 7. Etiquetado de la imagen miapp:v1 como miapp:latest.

1.4 Inspeccionar imagen

Se visualiza la estructura JSON devuelta por docker image inspect, donde aparecen campos de configuración de la imagen construida.

```
28 de mar 18:08
chaya@chaya-HP-240-G8-Notebook-PC: ~/docker-lab-chayanne
~/docker-lab-chayanne

            "Cmd": [
                "cat",
                "/app/mensaje.txt"
            ],
            "WorkingDir": "/app",
            "ArgsEscaped": true
        },
        "Architecture": "amd64",
        "Os": "linux",
        "Size": 3631069,
        "RootFS": {
            "Type": "layers",
            "Layers": [
                "sha256:f5b2c0456bf1080bb3f0617295df9721aa4aa27a536bba8e301c3af8d530c9cb",
                "sha256:f5d00d84879801470219bd0004e099620aeacf58c3fdf1c13837db32ef3b74a3",
                "sha256:469ef608fa0c14b2e16bed72cd78a207b3f460bcdad9b756d1dd3eb483d0153d"
            ]
        },
        "Metadata": {
            "LastTagTime": "2026-03-28T23:02:19.133766781Z"
        },
        "Descriptor": {
            "mediaType": "application/vnd.oci.image.index.v1+json",
            "digest": "sha256:67f2db8020c575ff247f8217f82eb50b5cfbcb95e4d4492087e724f6f52d761c",
            "size": 855
        },
        "Identity": {
            "Build": [
                {
                    "Ref": "1hilmog5tz1l5ird38pxih0em",
                    "CreatedAt": "2026-03-28T18:01:00.475211211-05:00"
                }
            ]
        }
    }
]
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 8. Inspección de la imagen miapp:v1.

1.4 a 1.7

En una sola evidencia se observan los comandos docker image inspect --format '{{.Id}}', docker save -o miapp.tar, docker load -i miapp.tar y docker rmi miapp:latest.

```
28 de mar 18:14
chaya@chaya-HP-240-G8-Notebook-PC: ~/docker-lab-chayanne
~/docker-lab-chayanne
                "sha256:469ef608fa0c14b2e16bed72cd78a207b3f460bcdad9b756d1dd3eb483d0153d"
            ]
        },
        "Metadata": {
            "LastTagTime": "2026-03-28T23:02:19.133766781Z"
        },
        "Descriptor": {
            "mediaType": "application/vnd.oci.image.index.v1+json",
            "digest": "sha256:67f2db8020c575ff247f8217f82eb50b5cfbcb95e4d4492087e724f6f52d761c",
            "size": 855
        },
        "Identity": {
            "Build": [
                {
                    "Ref": "1hilmog5tz1l5ird38pxih0em",
                    "CreatedAt": "2026-03-28T18:01:00.475211211-05:00"
                }
            ]
        }
    }
]
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker image inspect miapp:v1 --format '{{.Id}}'
sha256:67f2db8020c575ff247f8217f82eb50b5cfbcb95e4d4492087e724f6f52d761c
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker save -o miapp.tar miapp:v1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ ls -lh miapp.tar
-rw------- 1 chaya chaya 3.5M Mar 28 18:09 miapp.tar
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker rmi miapp:v1
Untagged: miapp:v1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker load -i miapp.tar
Loaded image: miapp:v1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:latest         67f2db8020c5       12.1MB         3.63MB
miapp:v1             67f2db8020c5       12.1MB         3.63MB
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker rmi miapp:latest
Untagged: miapp:latest
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 9. Obtención del ID, guardado a tar, carga desde tar y eliminación de etiqueta.

2.1 y 2.3

La captura muestra tanto el conflicto por reutilización del nombre web1 como la validación posterior mediante docker ps -a y docker ps, confirmando que el contenedor se encuentra en ejecución.

```
chaya@chaya-HP-240-G8-Notebook-PC: ~/docker-lab-chayanne
~/docker-lab-chayanne
chaya@chaya-HP-240-G8-Notebook-PC:~/Descargas$ CD
CD: no se encontró la orden
chaya@chaya-HP-240-G8-Notebook-PC:~/Descargas$ cd
chaya@chaya-HP-240-G8-Notebook-PC:~$ docker run -d --name web1 nginx:latest
11ab37613096e43e2d978f3afaf2255d154ecd2b1f86ab6c57dbd2e7c56d8207
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd Documentos
chaya@chaya-HP-240-G8-Notebook-PC:~/Documentos$ ls
 Automatizacion_playwright   Chayanne-Sanchez-  'const http require http.txt'
chaya@chaya-HP-240-G8-Notebook-PC:~/Documentos$ cd
chaya@chaya-HP-240-G8-Notebook-PC:~$ cd docker-lab-chayanne
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker run -d --name web1 nginx:latest
docker: Error response from daemon: Conflict. The container name "/web1" is already in use by container "11ab37613096e43e2d978f3afaf2255d154ecd2b1f86ab6c57dbd2e7
c56d8207". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS                     PORTS                                      NAMES
11ab37613096   nginx:latest   "/docker-entrypoint.…"   6 minutes ago   Up 6 minutes               80/tcp                                     web1
8155240381a1   nginx          "/docker-entrypoint.…"   5 days ago      Exited (255) 5 days ago    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp    mi-nginx
6ba0304c3179   hello-world    "/hello"                 5 days ago      Exited (0) 5 days ago                                                 compassionate_borg
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS     NAMES
11ab37613096   nginx:latest   "/docker-entrypoint.…"   9 minutes ago   Up 9 minutes   80/tcp    web1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 10. Ejecución del contenedor web1 y listado de contenedores.

2.2 Ejecutar con puertos

La respuesta HTML de nginx confirma que el servicio quedó expuesto correctamente en http://localhost:8080.

```
chaya@chaya-HP-240-G8-Notebook-PC: ~/docker-lab-chayanne
~/docker-lab-chayanne
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS     NAMES
11ab37613096   nginx:latest   "/docker-entrypoint.…"   9 minutes ago   Up 9 minutes   80/tcp    web1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker run -d --name web2 -p 8080:80 nginx:latest
038e1b01a619d1e9cdf392743e56d83880de5dd167e7490912b5762407152c74
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker ps
curl http://localhost:8080
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                       NAMES
038e1b01a619   nginx:latest   "/docker-entrypoint.…"   27 seconds ago   Up 23 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web2
11ab37613096   nginx:latest   "/docker-entrypoint.…"   11 minutes ago   Up 11 minutes   80/tcp                                      web1
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 11. Ejecución de web2 con mapeo de puerto 8080:80 y prueba con curl.

2.5 Inspeccionar contenedor

Se evidencia la consulta docker inspect web2 y parte del resultado JSON asociado a la configuración del contenedor.

```
chaya@chaya-HP-240-G8-Notebook-PC: ~/docker-lab-chayanne
~/docker-lab-chayanne
                "DriverOpts": null,
                "GwPriority": 0,
                "NetworkID": "e61dd0ab953fefafeed80e38862e978b6b28b670c7e7e099b179c62fef83c1e7",
                "EndpointID": "7f6bc309c1d3feeb648b7839276312ad6b4b054ad5274a5b57594af58f4cb462",
                "Gateway": "172.17.0.1",
                "IPAddress": "172.17.0.3",
                "MacAddress": "3a:75:db:92:74:ad",
                "IPPrefixLen": 16,
                "IPv6Gateway": "",
                "GlobalIPv6Address": "",
                "GlobalIPv6PrefixLen": 0,
                "DNSNames": null
            }
        }
    },
    "ImageManifestDescriptor": {
        "mediaType": "application/vnd.oci.image.manifest.v1+json",
        "digest": "sha256:c3fe1eeae810f4a585961f17339c93f0fb1c7c8d5c02c9181814f52bdd51961c",
        "size": 2290,
        "annotations": {
            "com.docker.official-images.bashbrew.arch": "amd64",
            "org.opencontainers.image.base.digest": "sha256:edc9450a9fe37d30b508808052f8d0564e3ed9eaf565e043c6f5632957f7381e",
            "org.opencontainers.image.base.name": "debian:trixie-slim",
            "org.opencontainers.image.created": "2026-03-24T22:11:24Z",
            "org.opencontainers.image.revision": "30c9ebf5aba124ca7bd79f97dd7c825c113ed864",
            "org.opencontainers.image.source": "https://github.com/nginx/docker-nginx.git#30c9ebf5aba124ca7bd79f97dd7c825c113ed864:mainline/debian",
            "org.opencontainers.image.url": "https://hub.docker.com/_/nginx",
            "org.opencontainers.image.version": "1.29.7"
        },
        "platform": {
            "architecture": "amd64",
            "os": "linux"
        }
    }
  }
]
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 12. Inspección detallada del contenedor web2.

2.5, 2.6 y 2.7

Se aprecia el uso de docker inspect --format '{{.State.Status}}', docker stop web2, docker start web2 y la sesión interactiva docker run -it --name alpine1 alpine:3.20 sh.

```
chaya@chaya-HP-240-G8-Notebook-PC: ~/docker-lab-chayanne
~/docker-lab-chayanne
            },
            "platform": {
                "architecture": "amd64",
                "os": "linux"
            }
        }
    }
]
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker inspect web2 --format '{{.State.Status}}'
running
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker stop web2
web2
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker start web2
web2
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED       STATUS          PORTS                                       NAMES
038e1b01a619   nginx:latest   "/docker-entrypoint.…"   2 hours ago   Up 18 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web2
11ab37613096   nginx:latest   "/docker-entrypoint.…"   2 hours ago   Up 2 hours      80/tcp                                      web1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ ^C
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker run -it --name alpine1 alpine:3.20 sh
Unable to find image 'alpine:3.20' locally
3.20: Pulling from library/alpine
Digest: sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805
Status: Downloaded newer image for alpine:3.20
/ # echo "prueba interactiva"
prueba interactiva
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.20.9
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ # exit
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker ps -a | grep alpine1
d1f7b1783318   alpine:3.20    "sh"                     About a minute ago   Exited (0) 45 seconds ago                                alpine1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 13. Validación del estado del contenedor, stop/start de web2 y modo interactivo con Alpine.

2.8 Pausar y reanudar

La captura muestra la secuencia docker pause web1, docker unpause web1 y la comprobación del estado mediante docker inspect web1 --format '{{.State.Paused}}'.

```
chaya@chaya-HP-240-G8-Notebook-PC: ~/docker-lab-chayanne
~/docker-lab-chayanne
        }
    }
]
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker inspect web2 --format '{{.State.Status}}'
running
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker stop web2
web2
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker start web2
web2
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED       STATUS          PORTS                                       NAMES
038e1b01a619   nginx:latest   "/docker-entrypoint.…"   2 hours ago   Up 18 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web2
11ab37613096   nginx:latest   "/docker-entrypoint.…"   2 hours ago   Up 2 hours      80/tcp                                      web1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ ^C
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker run -it --name alpine1 alpine:3.20 sh
Unable to find image 'alpine:3.20' locally
3.20: Pulling from library/alpine
Digest: sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805
Status: Downloaded newer image for alpine:3.20
/ # echo "prueba interactiva"
prueba interactiva
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.20.9
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ # exit
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker ps -a | grep alpine1
d1f7b1783318   alpine:3.20    "sh"                     About a minute ago   Exited (0) 45 seconds ago                                alpine1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker pause web1
web1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker unpause web1
web1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker inspect web1 --format '{{.State.Paused}}'
false
```

Figura 14. Pausa y reanudación del contenedor web1.

2.9 Eliminar contenedores

La salida final de docker ps -a evidencia que alpine1 y web1 fueron retirados, permaneciendo los contenedores esperados.

```
chaya@chaya-HP-240-G8-Notebook-PC: ~/docker-lab-chayanne
~/docker-lab-chayanne
Digest: sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805
Status: Downloaded newer image for alpine:3.20
/ # echo "prueba interactiva"
prueba interactiva
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.20.9
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ # exit
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker ps -a | grep alpine1
d1f7b1783318   alpine:3.20    "sh"                      About a minute ago   Exited (0) 45 seconds ago                                    alpine1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker pause web1
web1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker unpause web1
web1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker inspect web1 --format '{{.State.Paused}}'
false
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker inspect web1 --format '{{.State.Paused}}'
false
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker inspect web1 --format '{{.State.Paused}}'
false
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker stop alpine1
alpine1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
docker rm alpine1
alpine1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker rm -f web1
web1
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED       STATUS                     PORTS                                       NAMES
038e1b01a619   nginx:latest   "/docker-entrypoint.…"   2 hours ago   Up 20 minutes              0.0.0.0:8080->80/tcp, [::]:8080->80/tcp     web2
8155240381a1   nginx          "/docker-entrypoint.…"   6 days ago    Exited (255) 5 days ago    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp     mi-nginx
6ba0304c3179   hello-world    "/hello"                 6 days ago    Exited (0) 6 days ago                                                  compassionate_borg
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 15. Eliminación del contenedor detenido alpine1 y eliminación forzada de web1.

3.1 y 3.2

Se observa docker volume ls y docker volume inspect datos_nginx, incluyendo el punto de montaje del volumen.

```
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker volume ls
DRIVER    VOLUME NAME
local     datos_nginx
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker volume inspect datos_nginx
[
    {
        "CreatedAt": "2026-03-30T22:06:38-05:00",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/datos_nginx/_data",
        "Name": "datos_nginx",
        "Options": null,
        "Scope": "local"
    }
]
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 16. Listado e inspección del volumen datos_nginx.

3.3 Usar volumen

La captura muestra la creación de webvol, la copia del archivo index.html y la respuesta del servidor web con el texto personalizado.

```
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker run -d --name webvol -p 8081:80 -v datos_nginx:/usr/share/nginx/html nginx:latest
c518fb610a14ec69b0644f8960ee93029427e796bcf3924bef5d424a6d1372f0
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ echo "<h1>Hola volumen desde pc chayanne</h1>" > index.html
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker cp index.html webvol:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to webvol:/usr/share/nginx/html/index.html
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ curl http://localhost:8081
<h1>Hola volumen desde pc chayanne</h1>
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 17. Uso del volumen datos_nginx en un contenedor Nginx y validación con curl.

4.1 y 4.2

Además de docker network ls, se evidencia la creación de la red mi_red y la validación posterior del nuevo recurso.

```
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
e61dd0ab953f   bridge    bridge    local
c5ac45560933   host      host      local
eefa3547282c   none      null      local
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker network create mi_red
ed523c9b0a33fc60444fc2afb60dc4290e824de14fd512ea1c1b220b519feec0
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker netwok ls
docker: unknown command: docker netwok

Run 'docker --help' for more information
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
e61dd0ab953f   bridge    bridge    local
c5ac45560933   host      host      local
ed523c9b0a33   mi_red    bridge    local
eefa3547282c   none      null      local
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker network inspect mi_red
[
```

Figura 18. Listado de redes existentes y creación de la red bridge propia mi_red.

4.3 Inspeccionar red

La salida JSON muestra parámetros como nombre, identificador, driver, subnet y gateway de la red creada.

```
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker network inspect mi_red
[
    {
        "Name": "mi_red",
        "Id": "ed523c9b0a33fc60444fc2afb60dc4290e824de14fd512ea1c1b220b519feec0",
        "Created": "2026-03-30T22:53:01.39901165-05:00",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {},
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.18.0.0/16": {
                        "IPsInUse": 3,
```

Figura 19. Inspección de la red mi_red.

4.4 y 4.5

La evidencia complementaria muestra la creación de los contenedores caja1 y caja2 sobre la red mi_red, la resolución por nombre entre ambos mediante ping y, posteriormente, la desconexión de caja2 con el fallo esperado de la comunicación por nombre.

```
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker run -d --name caja1 --network mi_red alpine:3.20 sleep 300
1940afb087cf247407997811002 7d227dda74c426c570e3dc02fb7667bc09566
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker run -d --name caja2 --network mi_red alpine:3.20 sleep 300
d8f27f4dee9dd50bf0bff56e31f564096661a08695abf834b9cfd33d6b4d2e16
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker exec caja1 ping -c 2 caja2
PING caja2 (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=64 time=0.131 ms
64 bytes from 172.18.0.3: seq=1 ttl=64 time=0.215 ms

--- caja2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.131/0.173/0.215 ms
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker network disconnect mi_red caja2
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker exec caja1 ping -c 2 caja2
ping: bad address 'caja2'
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 20. Conectividad entre caja1 y caja2 en la red mi_red, seguida de la desconexión de caja2 y el fallo posterior del ping.

5. Docker Compose

Se incorporan las capturas faltantes del bloque Docker Compose, cubriendo la creación y corrección del archivo compose, el levantamiento del servicio, la revisión de logs, la baja del stack y el escalado del servicio web.

5.1 Crear y levantar servicios

La captura evidencia la corrección del archivo compose, la ejecución de docker compose up -d, la verificación con docker compose ps y la respuesta satisfactoria de Nginx en http://localhost:8082.

```
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ nano compose.yml
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker compose up -d
WARN[0000] Found multiple config files with supported names: /home/chaya/docker-lab-chayanne/compose.yaml, /home/chaya/docker-lab-chayanne/compose.yml
WARN[0000] Using /home/chaya/docker-lab-chayanne/compose.yaml
yaml: while parsing a block collection at line 4, column 7: line 4, column 18: did not find expected '-' indicator
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ rm compose.yaml
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ ls
DOckerfile  Dockerfile  compose.yml  index.html  miapp.tar
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker compose up -d
[+] up 2/2
 ✔ Network docker-lab-chayanne_default Created                                                                    1.5s
 ✔ Container docker-lab-chayanne-web-1 Started                                                                    7.0s
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker compose ps
curl http://localhost:8082
NAME                      IMAGE          COMMAND                  SERVICE   CREATED              STATUS          PORTS
docker-lab-chayanne-web-1   nginx:latest   "/docker-entrypoint.…"   web       About a minute ago   Up 57 seconds   0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>
```

Figura 21. Corrección del compose, levantamiento del servicio web y validación con docker compose ps y curl.

5.2 Revisar logs

La salida de docker compose logs confirma el arranque normal del servicio web y registra tanto solicitudes exitosas al sitio como el intento de acceso al favicon.

```
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker compose logs
web-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1  | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1  | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1  | 2026/03/31 04:38:08 [notice] 1#1: using the "epoll" event method
web-1  | 2026/03/31 04:38:08 [notice] 1#1: nginx/1.29.7
web-1  | 2026/03/31 04:38:08 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1  | 2026/03/31 04:38:08 [notice] 1#1: OS: Linux 6.17.0-14-generic
web-1  | 2026/03/31 04:38:08 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
web-1  | 2026/03/31 04:38:08 [notice] 1#1: start worker processes
web-1  | 2026/03/31 04:38:08 [notice] 1#1: start worker process 29
web-1  | 2026/03/31 04:38:08 [notice] 1#1: start worker process 30
web-1  | 2026/03/31 04:38:08 [notice] 1#1: start worker process 31
web-1  | 2026/03/31 04:38:08 [notice] 1#1: start worker process 32
web-1  | 172.19.0.1 - - [31/Mar/2026:04:38:59 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
web-1  | 172.19.0.1 - - [31/Mar/2026:04:39:09 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/
144.0.0.0 Safari/537.36" "-"
web-1  | 2026/03/31 04:39:10 [error] 30#30: *2 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.19.0.1, server: loca
lhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8082", referrer: "http://localhost:8082/"
web-1  | 172.19.0.1 - - [31/Mar/2026:04:39:10 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8082/" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/53
7.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36" "-"
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 22. Revisión de logs del servicio web mediante docker compose logs.

5.3 Bajar servicios

Se confirma la eliminación del contenedor y de la red creados por Compose mediante docker compose down, seguida de una consulta docker compose ps sin servicios activos.

```
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker compose down
[+] down 2/2
 ✔ Container docker-lab-chayanne-web-1 Removed                                    2.0s
 ✔ Network docker-lab-chayanne_default Removed                                    0.8s
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker compose ps
NAME      IMAGE     COMMAND   SERVICE   CREATED   STATUS    PORTS
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 23. Baja de servicios con docker compose down y verificación posterior con docker compose ps.

5.4 Escalar servicio

La evidencia incluye el error inicial por conflicto de puertos al escalar con publicación fija y el ajuste posterior que deja tres instancias activas del servicio web.

```
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker compose up -d --scale web=3
[+] up 3/4
 ✔ Network docker-lab-chayanne_default Created                                    0.5ss
 ✔ Container docker-lab-chayanne-web-1 Started                                    10.1s
 ✔ Container docker-lab-chayanne-web-3 Created                                    7.8ss
 ⠿ Container docker-lab-chayanne-web-2 Starting                                   14.6s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint docker-lab-chayanne-web-2 (0d2dd88
eda32fb891e617bf7e2b5752877ce8bb52cc6d4f79438fc1ff8129fcd): Bind for 0.0.0.0:8082 failed: port is already allocated
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker compose down
[+] down 4/4
 ✔ Container docker-lab-chayanne-web-3 Removed                                    1.1s
 ✔ Container docker-lab-chayanne-web-1 Removed                                    4.4s
 ✔ Container docker-lab-chayanne-web-2 Removed                                    1.0s
 ✔ Network docker-lab-chayanne_default Removed                                    0.5s
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ nano compose.yml
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker compose up -d --scale web=3
[+] up 4/4
 ✔ Network docker-lab-chayanne_default Created                                    0.2ss
 ✔ Container docker-lab-chayanne-web-3 Started                                    9.8ss
 ✔ Container docker-lab-chayanne-web-1 Started                                    6.6ss
 ✔ Container docker-lab-chayanne-web-2 Started                                    12.4s
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$ docker compose ps
NAME                      IMAGE          COMMAND                  SERVICE   CREATED          STATUS          PORTS
docker-lab-chayanne-web-1 nginx:latest   "/docker-entrypoint.…"   web       52 seconds ago   Up 46 seconds   80/tcp
docker-lab-chayanne-web-2 nginx:latest   "/docker-entrypoint.…"   web       52 seconds ago   Up 40 seconds   80/tcp
docker-lab-chayanne-web-3 nginx:latest   "/docker-entrypoint.…"   web       52 seconds ago   Up 43 seconds   80/tcp
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne$
```

Figura 24. Escalado del servicio web con --scale web=3, incluyendo el ajuste posterior para dejar tres instancias en ejecución.

6. Dockerfile (instrucciones)

Finalmente se incorpora la evidencia del ejemplo corto del bloque 6 de la guía, orientado a validar instrucciones como ARG, ENV, LABEL, ADD/COPY, VOLUME y CMD dentro de una imagen de prueba.

6.1 Ejemplo corto y ejecución

El build de dockerfile-demo con --build-arg VERSION=2.0 finaliza correctamente y docker run --rm dockerfile-demo imprime APP_ENV, el contenido de archivo.txt y la versión construida.

```
 => => unpacking to docker.io/library/dockerfile-demo:latest                      0.0s
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne/dockerfile-test$ docker build -t dockerfile-demo --build-arg VERSION=2.0 .
[+] Building 4.5s (10/10) FINISHED                                               docker:default
 => [internal] load build definition from Dockerfile                              0.2s
 => => transferring dockerfile: 344B                                              0.0s
 => [internal] load metadata for docker.io/library/alpine:3.20                    0.3s
 => [internal] load .dockerignore                                                 0.1s
 => => transferring context: 2B                                                   0.0s
 => [1/5] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805  0.3s
 => => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805  0.3s
 => [internal] load build context                                                 0.2s
 => => transferring context: 32B                                                  0.0s
 => CACHED [2/5] WORKDIR /app                                                     0.0s
 => CACHED [3/5] COPY archivo.txt /app/archivo.txt                                0.0s
 => CACHED [4/5] RUN echo "version=2.0" > /app/version.txt                        0.0s
 => CACHED [5/5] ADD archivo.txt /app/archivo2.txt                                0.0s
 => exporting to image                                                            2.1s
 => => exporting layers                                                           0.0s
 => => exporting manifest sha256:6996e147a564b2828724341500cc196e496777dd39dc9d97f5e8a6687a50ff66  0.0s
 => => exporting config sha256:58f3fed395fbb371778932d9929a8ec446bb94c77a72a33f188fd109022f9401  0.0s
 => => exporting attestation manifest sha256:e2cd833362272e496a8b2b76cf9d41d8bc9e93299c4e0f36e6eebe34a9031f2f  0.4s
 => => exporting manifest list sha256:90caefcfceeb3c9caa8f642e4eb6318dd3ad0528ea6c1c0312e8bc530eb1f816  1.1s
 => => naming to docker.io/library/dockerfile-demo:latest                         0.1s
 => => unpacking to docker.io/library/dockerfile-demo:latest                      0.1s
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne/dockerfile-test$ docker run --rm dockerfile-demo
APP_ENV=dev
echo "contenido local"
version=2.0
```

Figura 25. Construcción de dockerfile-demo con build-arg VERSION=2.0 y validación de su ejecución.

6.2 Inspección de la imagen

La salida de docker image inspect dockerfile-demo confirma la presencia del LABEL autor, la variable APP_ENV=dev, el CMD configurado y el VOLUME /data.

```
chaya@chaya-HP-240-G8-Notebook-PC: ~/docker-lab-chayanne/dockerfile-test
~/docker-lab-chayanne/dockerfile-test
chaya@chaya-HP-240-G8-Notebook-PC:~/docker-lab-chayanne/dockerfile-test$ docker image inspect dockerfile-demo
[
    {
        "Id": "sha256:90caefcfceeb3c9caa8f642e4eb6318dd3ad0528ea6c1c0312e8bc530eb1f816",
        "RepoTags": [
            "dockerfile-demo:latest"
        ],
        "RepoDigests": [
            "dockerfile-demo@sha256:90caefcfceeb3c9caa8f642e4eb6318dd3ad0528ea6c1c0312e8bc530eb1f816"
        ],
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-03-31T00:24:04.768311463-05:00",
        "Config": {
            "Env": [
                "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
                "APP_ENV=dev"
            ],
            "Cmd": [
                "sh",
                "-c",
                "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat /app/version.txt"
            ],
            "Volumes": {
                "/data": {}
            },
            "WorkingDir": "/app",
            "Labels": {
                "autor": "prueba"
            },
            "ArgsEscaped": true
        },
        "Architecture": "amd64",
        "Os": "linux",
        "Size": 3632977,
        "RootFS": {
            "Type": "layers",
            "Layers": [
                "sha256:f5b2c0456bf1080bb3f0617295df9721aa4aa27a536bba8e301c3af8d530c9cb",
```

Figura 26. Inspección de la imagen dockerfile-demo para verificar metadatos, variables y configuración de ejecución.

7. Conclusiones

Con la incorporación de las capturas complementarias, el informe cubre todos los bloques principales planteados en la guía del laboratorio: preparación, imágenes, contenedores, volúmenes, redes, Docker Compose y validación de instrucciones de Dockerfile.

Las evidencias permiten seguir una secuencia verificable de comandos y resultados, desde la instalación y descarga de imágenes hasta la orquestación básica con Compose y la inspección de metadatos de imágenes personalizadas.

El documento queda así en condición de soporte integral para la revisión del docente, conservando el formato del informe inicial y ampliando la cobertura visual con las nuevas pruebas suministradas.